

Assignment No. 1

Association Rule Learning – Market Basket Analysis

Problem Description

The sales transactions of a retail store over a specific time period are provided in the file **factors.csv**. The objective of this assignment is to perform **Association Rule Learning** and analyze customer purchasing behavior using market basket analysis techniques.

Dataset Attributes

Attribute	Description
Order_id	Invoice number
Product_id	Unique product identifier
Product_name	Product name
Department_id	Department identifier
Department_name	Department name

Part A – Global Association Analysis

As a data analyst, you are required to generate the following reports:

- Identify which products are frequently purchased together by customers.
- Identify which store departments commonly appear together in customers' shopping baskets.

Each report must include:

- Names of product pairs (or department pairs in the second report)
- Support value
- Confidence value

Part B – Department-Based Analysis

After some time, the store decides to manage each department independently. You are now required to generate the following report separately for each department:

- Identify which products are frequently purchased together within each department.

Association Rule Constraints

Use the following thresholds for all generated association rules:

Parameter	Value
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Minimum Support	0.3% (min_support = 0.003)
Minimum Confidence	25% (min_confidence = 0.25)

Additional Notes

- A sample output for Report No. 1 using the first 50% of the dataset is provided in the file report.csv.
- The sample report contains only product codes.
- Managers require both product codes and product names in the final reports.